

AI Engineer Foundation

Python for Data Science & Mathematics for AI

Mini-Project	Analyzing patient data, building mathematical insights, and processing data to support the AI pipeline
Time	2 weeks
Start	05/05/2026
End	15/05/2026

Dataset: Heart Failure Prediction

URL: [Heart Failure Prediction Dataset](#)

Dataset Description

The dataset includes the following key information: the target variable is **HeartDisease** (predicting the presence of heart disease: 1 for presence, 0 for normal). Clinical features include categorical variables such as **Sex**, **ChestPainType**, **RestingECG** result, **ExerciseAngina** (exercise-induced angina), and the slope of the ST segment (**ST_Slope**). Additionally, there are numerical variables representing specific health indices such as **Age**, **RestingBP** (resting blood pressure), **Cholesterol** (serum cholesterol), **FastingBS** (fasting blood sugar), **MaxHR** (maximum heart rate achieved), and the ST segment depression level (**Oldpeak**).

This dataset is the largest combined version (918 observations) from 5 popular medical datasets (Cleveland, Hungarian, Switzerland, Long Beach VA, Stalog) and is commonly used to explore Machine Learning models for early detection and management of cardiovascular disease.

Questions

MODULE 1: NUMPY (10 Questions)

Q1. Load the dataset into a NumPy array **using only the numerical features** (e.g., Age, RestingBP, Cholesterol, MaxHR, Oldpeak). Check the shape and data type (dtype) of the array.

Q2. Calculate the Mean, Median, and Standard Deviation for the Age, RestingBP, and Cholesterol columns using NumPy functions.

Q3. Apply Min-Max normalization to scale all numerical features to the range [0, 1] using vectorized operations.

Q4. Use boolean indexing to filter out patients with Age > 50 and a Cholesterol index greater than the average of the entire dataset.

Q5. Calculate the correlation matrix between the numerical features entirely with NumPy (without using Pandas).

Vectorization & Linear Algebra

Q6.: Write a fully vectorized function to calculate the Euclidean distance between patients using only numerical features (avoiding standard Python loops).

Q7. Implement from scratch both Min-Max and Z-score normalization methods using NumPy and compare their results.

Q8. Calculate the Cosine similarity between two selected patient vectors from the dataset (using only numerical features).

Q9. Manually perform the PCA (Principal Component Analysis) algorithm using NumPy: construct the covariance matrix from the numerical variables, calculate the eigenvalues and eigenvectors, and then project the data onto the top 2 principal components.

Q10. Implement batch processing by dividing a NumPy array into equally sized batches and calculating the mean vector for each batch.

MODULE 2: MATHEMATICS FOR AI (10 Questions)

Q1. Represent the dataset as a numerical matrix (using only numerical features) and determine the shape and rank of the matrix. X

Q2. Using the target variable HeartDisease, apply Linear Regression using Age and MaxHR to

predict heart disease based on the model: $y = Xw + b$

Q3. Calculate the basic probabilities $P(\text{HeartDisease} = 1)$ and conditional probabilities $P(\text{HeartDisease} = 1 \mid \text{Sex} = \text{M})$ from the dataset.

Q4. Based on the Mean Squared Error (MSE) loss function, prove the derivative formula and calculate the Gradient with respect to the weights w for the dataset: $L = \frac{1}{n} \sum (y - \hat{y})^2$

Q5. Calculate the variance of Cholesterol and the covariance between Cholesterol and Age.

Q6. Construct the covariance matrix using all numerical features and calculate its eigenvalues and eigenvectors.

Q7. Perform Singular Value Decomposition (SVD) on the matrix X (consisting only of numerical variables) and explain the meaning of the matrices U , Σ , and V^T and in the context of patient profile analysis: $X = U\Sigma V^T$

Q8. Implement the Gradient Descent algorithm to optimize the parameters of the Linear Regression model in Q2 and visualize the convergence of the loss function over iterations.

Q9. Discretize the Age variable into groups (e.g., Young/Middle-aged/Senior), then apply Bayes' Theorem to calculate $P(\text{HeartDisease} \mid \text{Age_bin})$

Q10. Apply both L1 (Lasso) and L2 (Ridge) regularization techniques to the Linear Regression model and compare their impact on the model coefficients and predictive performance.

MODULE 3: PANDAS (10 Questions)

Q1. Load the dataset into a Pandas DataFrame and display head(), info(), and describe() to explore the structure of the 12 attributes (comprising 918 rows).

Q2. Clean the data: handle invalid values (e.g., check for cases where Cholesterol or RestingBP equals 0 and replace with NaN/Mean if necessary) and ensure the correct data type for all columns.

Q3. Group the data (Groupby) by the ChestPainType variable and calculate the total number of patients as well as the Heart Disease rate for each group.

Q4. Create a new feature MaxHR_to_Age_Ratio = MaxHR / Age to analyze the level of exertion

relative to age.

Q5. Visualize the distribution of Cholesterol using a distribution plot (histogram) and compare the number of patients with/without heart disease using a bar chart.

Q6. Create a pivot table with RestingECG as the index, HeartDisease as columns, and the count of patients as the values.

Q7. Create categorical groups based on age Age_Group (e.g., <40, 40-55, 55+ years) and analyze the heart disease rate across these age groups.

Q8. Create a new categorical feature based on Cholesterol using the binning technique (e.g., Normal, Borderline, High) and analyze the behavior of the disease across each segment.

Q9. Create advanced visualizations: a correlation heatmap for the numerical features and a boxplot of MaxHR grouped by HeartDisease.

Q10. Perform Feature Engineering: conduct One-Hot Encoding on the categorical variables (Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope) to prepare a clean, complete dataset, ready to be fed into machine learning models.

Submission Requirements and Implementation Guidelines

Students are required to complete the mini-project using Python, implemented either in LMS Quanskill, VS Code (local environment), or Google Colab (cloud environment). The submission must demonstrate a complete and reproducible workflow from data loading to analysis and results.

Each submission should include: a well-structured Python script or notebook (.py or .ipynb) with clear organization of code into logical sections (e.g., data loading, preprocessing, analysis, visualization); appropriate use of libraries such as NumPy, Pandas, and Matplotlib/Seaborn; and meaningful comments explaining key steps and decisions.

Students must ensure that the code runs without errors and produces all required outputs (tables, statistics, and visualizations). In addition, a brief report (or markdown cells in Colab) should summarize the main findings, insights, and conclusions derived from the dataset.

All files should be submitted in a single package, including the source code and any necessary supporting files or instructions to reproduce the results.

Example Directory Structure

heart-failure-project/

```
|
|— data/
|   |— raw/      # Original, immutable data files
|   |   |— heart.csv
|   |— processed/ # Intermediate or final data files (e.g., after cleaning/pre-processing)
|
|— notebooks/    # Jupyter notebooks for exploration and analysis
|   |— exploratory_analysis.ipynb
|
|— src/          # Source code for the project (modules, classes, functions)
|   |— data_processing.py
|   |— feature_engineering.py
|   |— modelling.py
|   |— utils.py
|
|— outputs/      # Generated output from the analysis
|   |— figures/   # Generated plots and images
|   |— tables/    # Generated tables
|   |— metrics.txt # Key model metrics/results (renamed from results.txt for clarity)
|
|— report/       # Final project documentation
|   |— final_report.pdf
|
|— requirements.txt # Project dependencies
|— README.md       # Project overview and setup instructions
```

